

Notes

AI will not become AGI without a couple of changes in the paradigm. In other words, the current transformer architecture will not be enough even if more and more data are added and more compute is added, as sometimes is promised by many adventurous AI futurists.

Rather, what is needed according to Yann LeCun are persistent memory, the ability to learn from the real world, and the ability to plan. For those characteristics to materialize, AI will probably need to start learning from videos and build world models from those observations of the videos. That is similar to how small children learn how the world works.

Another thought is that the last mile is the hardest for each new technology that is being deployed at scale. A good example are self-driving cars, level five is much harder to achieve than getting from level three to level four, because you need complete reliability. With artificial intelligence we see that in the case of deep research: if 95% of the results are correct and we don't know what 5% are hallucinations, then this still calls for human judgment and expertise to make productive use of these systems. It's notable that these evaluations can be automated in some cases, like we have seen in the case of AlphaEvolve, where code can be evaluated automatically and put into formal proof systems. In those cases, the ability of the artificial intelligence to generate ideas or solution candidates in combination with the evaluation, does generate a reliable system. By and large, we are still at the stage where a human judgment and the human in the loop is critical.